Kamogelo Mmopane

Full Stack Developer – React, TypeScript, Node.js, PostgreSQL

Johannesburg, South Africa

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I'm a **Full Stack Developer** specialising in **React, Node.js,** and **TypeScript**. I take full ownership of the end-to-end pipeline: architecting secure **RESTful APIs** with **Express.js**, managing **PostgreSQL** databases alongside **Supabase, MongoDB,** and **Prisma ORM**, optimizing performance with **Redis** caching, and containerising with **Docker**.

I build responsive, component-driven interfaces using **Next.js, Redux, JavaScript,** and **Tailwind CSS**, testing rigorously with **Jest/Vitest**, and automating delivery through **CI/CD** pipelines. My current areas of interest include cloud architecture, with a focus on **Microsoft Azure**. I have spent the last **2 years** shipping functional web software and internal tools for tech start-ups.

Projects

OCCI (Code | View Site)

March - June 2026

A modern **Next.js** and **TypeScript** platform engineered for an ICT infrastructure firm to automate client onboarding pipelines and communication flows.

- Built a mobile-first, responsive frontend from scratch using **Next.js, TypeScript,** and **Tailwind CSS** to optimize cross-device client data capture.
- Configured real-time server-side routing mechanisms to instantly process inbound client queries.
- Integrated automated lead-capture workflows with dual notifications to route client validation updates securely.

E-Commerce Engine (Code | View Site)

January - March 2025

A full-stack **React, JavaScript, Node.js, Express.js** commercial engine running scalable database transactions, automated payment configurations, and rigid main-branch delivery constraints.

- Engineered the core relational database schema utilizing **PostgreSQL and Prisma ORM** to support automated checkout structures.
- Packaged the entire commercial application into isolated, deployable build environments using **Docker**.
- Built and executed an automated verification pipeline using **Jest** to ensure zero-regression code delivery.

Experience

Powerverse - Front End Developer (React, TypeScript, Next.js, Docker, Vitest)

May 2026 - Present

Delivering production-ready client interfaces within cross-functional Agile engineering teams for high-priority public sector data systems.

- Built a digital multi-step workflow platform using React, TypeScript, and Tailwind CSS that digitized manual Department of Transport paper operations.
- Developed asynchronous data synchronization logic and frontend rendering tests using Vitest to align client application state with Docker environments.
- Programmed modular frontend component layouts that integrated with asynchronous REST API endpoints to securely process massive government datasets.

KeelWorks Foundation - Full Stack Developer (React, Node.js, Express.js, MongoDB)
March - September 2024

Engineered a secure user authentication system for a Portfolio Generator platform inside a global, distributed Agile team for the Tech Non-Profit industry.

- Coded secure backend middleware endpoints using Node.js and Express.js to handle authentication workflows for over 100 live users.
- Integrated custom token interceptors via Axios on React to catch session timeouts and securely handle corrupted application tokens.
- Developed a token-driven user session configuration system to handle encrypted client accounts across a distributed MongoDB database layer.

Zaio Institute of Technology - Full Stack Developer (React, Redux, Node.js, Express.js, MongoDB)
June 2022 - July 2023

Constructing a high-performance note-taking engine for an E-Learning platform, pairing up with backend teams to optimize live application speeds.

- Fixed query performance bottlenecks by implementing Mongoose pagination, frontend lazy loading, and targeted database indexing schemas across MongoDB.
- Integrated a responsive React and JavaScript note-taking module with backend Express.js APIs backed by global Redux state logic.
- Restructured application database schemas to cleanly decouple student profiles from active user components and maximize application data load speeds.

Education

Occupational Certificate: Software Engineer
WeThinkCode_, South Africa,
September 2024 - December 2025

- Python and Java Fundamentals
- Object-Oriented Programming (OOP)
- CI/CD, Docker, Git and Linux
- SQL, Relational Databases and ORMs
- Unit, Integration and Acceptance Testing
- HTML5, CSS3 and JavaScript
- Service-Oriented Architecture, Message Queues

Full Stack Web Development
Zaio Institute of Technology, South Africa,
February 2022 - November 2023

National Senior Certificate
Moletsane High School, South Africa,
January 2005 - November 2011